Disaster Evacuation & Emergency Response Simulation

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An Agent-Based GIS Approach for Santa Cruz, Laguna

Project Objectives

Primary Goal: To simulate and analyze evacuation efficiency during a localized fire emergency.

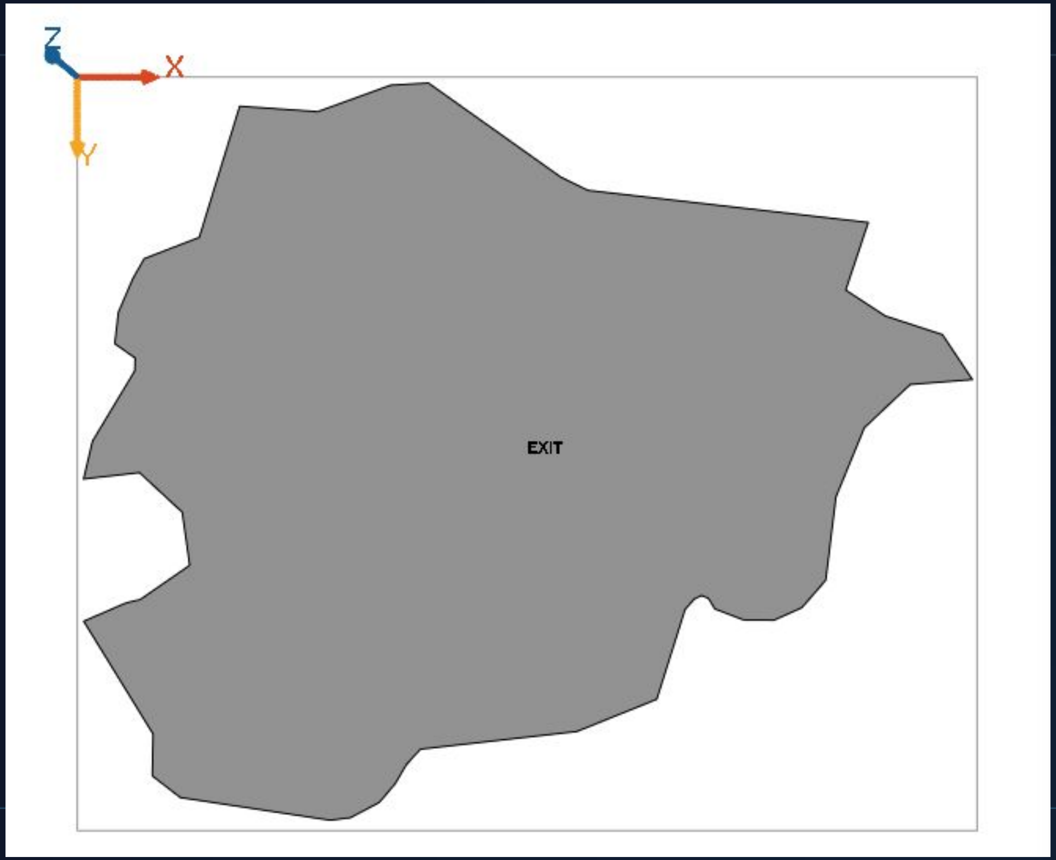
- **Identify Bottlenecks:** Pinpoint physical constraints within the GIS environment.
- **Measure Velocity Impact:** Analyze how randomized speeds affect total evacuation time.
- **Evaluate Safe Zones:** Visualize the effectiveness of dedicated assembly points.

The Core Challenge

Understanding how civilian populations react spatially to emergency stimuli within accurate municipal footprints.

Simulation Environment (GIS)

- **Data Source:** Vector-based shapefiles (.shp) of local structural footprints.
- **Obstacles (Gray Polygons):** Represent static building boundaries constraints.
- **Exit Point (Green Marker):** Designated Safe Zone for the simulated population.
- **Spatial Realism:** Real-world physical constraints dictate pathfinding limitations.



Agent-Based Modeling (ABM)

Heterogeneity

Agents possess unique, randomized speeds.

$v \in [3.0, 7.0]$ m/s

Autonomy

Agents independently calculate their "goto" vectors based on real-time distance to the designated exit.

Emergent Behavior

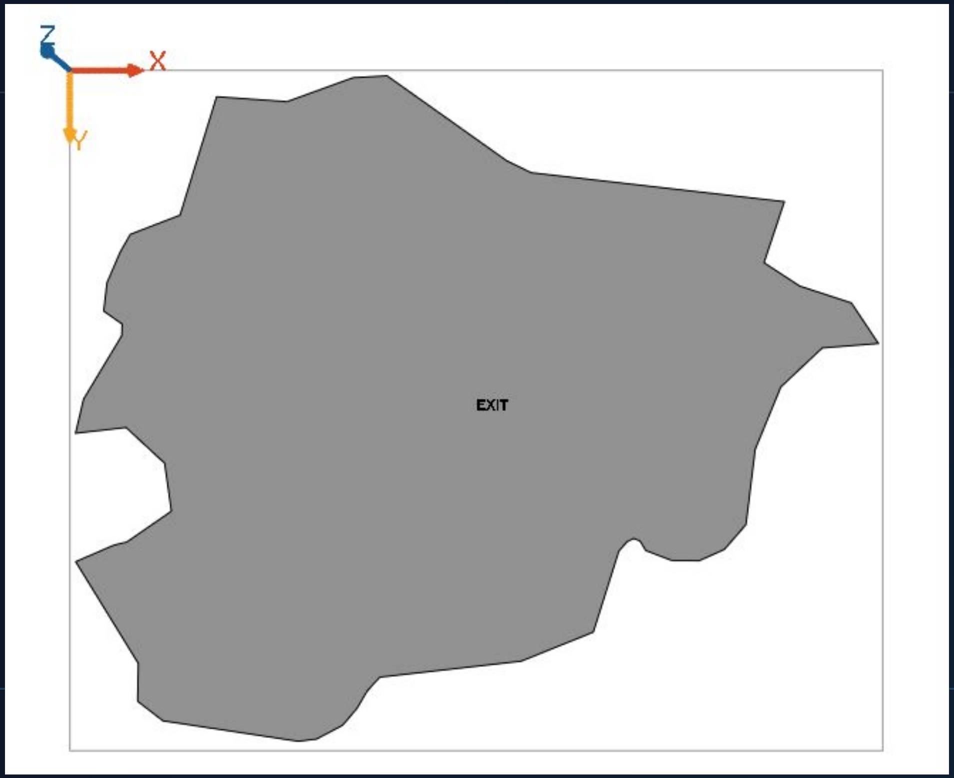
Individual, uncoordinated escape attempts collectively manifest as a "convergent flow" pattern.

Agent Behaviors & Logic

Phase	Agent Action
Initialization	Spawned at random coordinates within map boundaries.
Navigation	Executes goto command for goal-oriented pathing.
Pathfinding	Moves in a direct vector, constrained by spatial bounds.
Termination	At < 3m from exit, increments "Escaped" counter and is removed.

Visualization Techniques

- **Movement Trails:** Facet tracing visualizes path history, highlighting primary evacuation corridors.
- **Dynamic States:** High-visibility red coloring signals active emergency status to the user.
- **Directional Vectors:** Solid indicator lines from agents to the exit clarify intent.

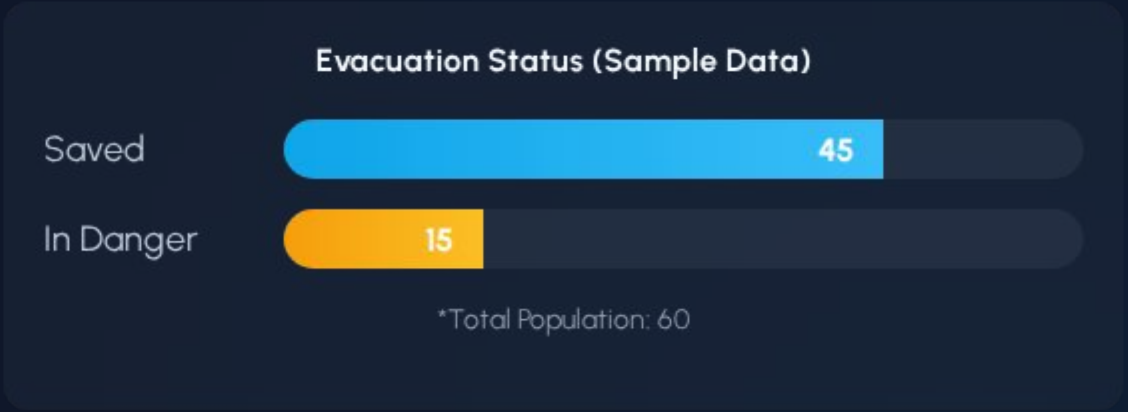


Analytics & Monitoring

Real-Time Dashboards
Live tracking of "Remaining Citizens" vs. "Total Saved".

Time-Series Analysis
Line charts map the decline of at-risk population over time.

>_Console Logging
Cycle-by-cycle text outputs for granular speed tracking.



Simulation Results

Convergent Flow

Agents naturally formed concentrated streams toward the primary exit point, validating pathfinding logic.

Density Issues

Significant traffic deceleration occurred near the exit coordinate, creating measurable "choke points".

Clearance Time

Successfully established baseline clearance metrics for a 60-person cohort within this specific layout.

Future Improvements

- **Obstacle-Aware Pathfinding:**
  Implement non-linear navigation graphs (e.g., A* algorithm) to prevent clipping through building polygons.
- **Dynamic Hazard Simulation:**
  Introduce a spreading "Fire Agent" that dynamically forces civilian agents to recalculate escape routes.
- **Heterogeneous Populations:**
  Model specific demographic behaviors, including slower moving entities (elderly/children) and inbound First Responders.

Conclusion

GAMA and GIS integration effectively predicts critical choke points in emergency scenarios.

Key Takeaway: These digital twins empower local authorities to proactively plan safer infrastructure and optimize evacuation protocols for the Santa Cruz community.